



Wind-driven shear stress on sediment phosphorus release: Implications for water quality, and harmful algal blooms in a shallow estuary

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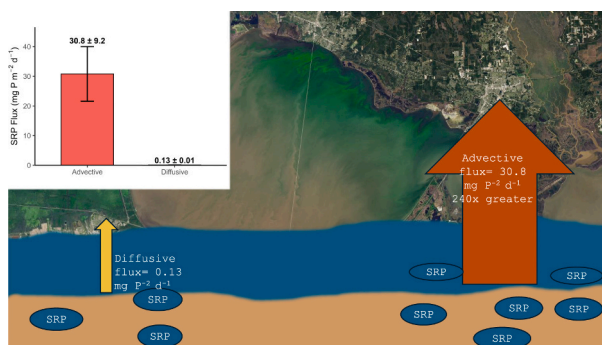
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HIGHLIGHTS

- Advective P flux increased linearly with increasing shear stress, representative of mild to intense wind/storm conditions.
- Advective P flux was ~240 times greater than diffusive P flux from estuarine sediment.
- Storms lead to increased water column P release from sediment.

GRAPHICAL ABSTRACT



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ABSTRACT

Internal phosphorus loading from sediments can act as a persistent nutrient source that fuels eutrophic conditions and harmful algal blooms (HABs), two global water quality issues. While most studies estimate the concentration gradient-driven diffusive flux, the role of wind-driven sediment disturbance remains less explored in shallow, coastal systems. This study quantified soluble reactive phosphorus (SRP) flux from sediments in a shallow subtropical estuary, quantifying both the diffusive and advective pathways. Sediment mapping at 160 sites identified strong west-east gradients in total phosphorus (TP), with four representative sites selected for laboratory incubations. Aerobic diffusive flux was measured over two weeks in intact cores, while advective flux was determined under increasing bedload shear stress (0.01–0.45 Pa) at 20-minute intervals using a Gust Erosion Microcosm. Diffusive SRP flux averaged $0.13 \text{ mg P m}^{-2} \text{ d}^{-1}$ ($\sim 0.21 \text{ Mt d}^{-1}$ estuary-wide) and was below $0 \text{ mg P m}^{-2} \text{ d}^{-1}$ at half the sites, markedly lower than previous estimates. Advective flux averaged $30.8 \text{ mg P m}^{-2} \text{ d}^{-1}$ ($\sim 49.3 \text{ Mt d}^{-1}$), representing up to a ~ 240 -fold increase over diffusive flux. The advective flux scaled linearly with shear stress, suggesting that even moderate winds can mobilize substantial sediment SRP pools. Seasonal autumn and winter cold fronts may deplete sediment SRP, reducing availability during summer bloom-favorable conditions. In contrast, tropical storms in the summer months could trigger significant P releases under optimal HAB growth conditions. These findings underscore the importance of incorporating episodic, wind-driven

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1. Introduction

Globally, nutrient loading of estuarine systems is regarded as one of the major contributors to water quality degradation (Cullen, 1986; Smith, 2003). Excess nutrient loading into aquatic systems contributes to eutrophication by enhancing primary production and the subsequent accumulation of organic matter in the water column (Pinckney et al., 2001; Nixon, 1995). The nutrients then become concentrated in the sediment, where conditions may lead to subsequent release, frequently referred to in the literature as the internal nutrient load (Reddy et al., 2011; Wang et al., 2006; Immers et al., 2015).

Mitigating nutrient loading from external sources has been a key focus for watershed managers to combat increasing rates of eutrophication and degraded water quality (Jickells, 2005; Chambers et al., 2006). Particular focus has been aimed at controlling and mitigating excess external P loading to coastal systems (Wurtsbaugh et al., 2019). Once externally loaded, the effect of P can compound due to its remaining in a closed loop between the sediment and the water column after introduction to the system, and consequently contributing to the fueling of nitrogen-fixing HABs (Søndergaard et al., 2003; Bargu et al., 2023).

Movement of P below the sediment-water interface (anaerobic conditions) to the aerobic water column occurs primarily through two internal loading pathways: diffusion of dissolved P and advective flux following a disturbance of the sediment-water interface (Reddy et al., 1996; Wang et al., 2003). Diffusive flux occurs due to a concentration gradient between the sediment pore space and the overlying water column. In contrast, the advective flux of P occurs through disturbances to the sediment layer caused by bioturbation, wind, waves, or tidal stress (Wang et al., 2024). These disturbances release loosely consolidated surface sediments into the water column, increasing turbidity (You et al., 2007; Reddy et al., 1996). Perturbation of the sediment-water interface consequently results in the release of nutrients, which accumulate in the anaerobic sediment layer in soluble form (Boynton et al., 2018). Once released, these nutrients flux into the water column, where they remain suspended until they bind to minerals in the aerobic oxidized water column or are taken up by primary producers (Reddy et al., 1996). The SRP released into the water column can be assimilated by phytoplankton, including harmful algal species (Cymbola et al., 2008). Other potential fates for the released P include sorption onto iron oxides, precipitation with Ca-Mg-based minerals, or export to the coastal ocean (Gu et al., 2024; Adhikari et al., 2015). Under anoxic water-column conditions, bound P can be released at a more rapid rate due to the reduction of the upper sediment layer, which typically retains P through sorption to oxidized iron (Reddy et al., 2022).

The internal loading of P can fuel harmful algal blooms (HABs), particularly in areas with P-limiting conditions or where N-fixing HABs are prevalent, further deteriorating the system's environmental quality (Anderson et al., 2008). These HABs are an issue along much of the U.S. coastlines (Anderson et al., 2021) as well as across the globe (Hallegraeff et al., 2021). The occurrence of HABs in coastal areas leads to hypoxia, fish kills, and harm to the local human population, making them an increasing concern for coastal managers (Heil and Muni-Morgan, 2021; Hallett et al., 2016).

Coastal Louisiana is particularly vulnerable to eutrophication, in part due to nutrient inputs from the Mississippi River (Turner and Rabalais, 1994). The Mississippi River drains approximately 42 % of the continental United States and carries significant fertilizer runoff and wastewater from municipalities along its course (Turner, 2024). The Lake Pontchartrain Estuary has experienced HABs in response to the opening of the Bonnet Carré Spillway, a flood release valve for the Mississippi

River (Snow et al., 2024; Bargu et al., 2023). Along with freshwater input, spillway operation results in nutrient loading from the Mississippi River into the estuarine system, with the majority of P in particulate form (Roy et al., 2012, 2017; McCorquodale et al., 2009; Zhang et al., 2012). Diffusive flux mechanisms governing the internal loading of P into the water column following an opening have been the focus of recent studies (Roy et al., 2017); however, no studies have measured the advective flux, driven by the wind and wave energy imparted upon the sediment.

Seasonality can have a significant impact on the timing and duration of wind patterns in coastal systems, as well as the subsequent wind-induced mixing of surface sediments (Stone and Deegan, 1980; Hiatt et al., 2019). Across the Gulf Coast, winter months are characterized by regular cold fronts that introduce consistent, strong winds to the system (Hiatt et al., 2019). The low diurnal tidal prism has a minimal effect on sediment-bed shear stress in this system compared to systems like the Chesapeake Bay with larger tidal regimes (McCorquodale et al., 2009; Li et al., 2022).

Consequently, winds over the surface of shallow waterbodies are the dominant force contributing to shear stress across the sediment bed. This force disturbs the top sediment layer and releases porewater nutrients directly into the water column (Stone and Deegan, 1980; Li et al., 2008). This disturbance of the sediment is persistent in the fall and winter months along the Gulf coast, where cold fronts result in winds of 8–12 m s⁻¹ every 3–7 days (Hiatt et al., 2019; Huang and Li, 2019). In addition to winter cold fronts, hurricane seasons between June and November along the US Gulf and East Coasts can result in powerful storms that introduce severe winds and heavy precipitation. These events can both increase external loading from the watershed and the advective release of sediment-bound P internally (Roth, 1998). However, only a few empirical studies have measured the release rate of SRP into the water column following the application of shear stress to the sediment. Kleeberg et al. (2007) examined P release from river flow, while Li et al. (2022) examined P release due to tidal forces in the Chesapeake Bay. Only one study compared diffusive and advective release of P in a hypereutrophic lake, although the advective flux was not calibrated to any known shear force (Reddy et al., 1996). With these limited studies in mind, this study aimed to determine the dominant internal P loading mechanism from the Lake Pontchartrain Estuary by 1) mapping the spatial distribution of P in the estuarine sediments, and using this distribution, measuring 2) the diffusive flux rate of SRP and 3) the advective SRP flux rates over a range of bed load shear stresses. Our hypothesis states that wind-induced shear stress from storm events would serve as the primary mechanism for SRP release to the water column in the Lake Pontchartrain Estuary. Additionally, we aimed to compare these shear stresses to wind speeds to predict how weather conditions contribute to internal P loading. This approach can be duplicated in other regions struggling with P-induced HABs to allow coastal resource managers to determine the dominant process for P release from the sediment, which can lead remediation efforts.

2. Materials and methods

2.1. Site description

The Lake Pontchartrain Estuary (LPE) is a shallow (~3.7 m) oligotrophic to mesotrophic estuary in southeastern Louisiana, north of New Orleans, with an average volume of ~6 km³ (Turner et al., 2002) (Fig. 1). The estuary has a surface area of roughly 1620 km², a north-south axis of 40.2 km, and an east-west axis of 64.4 km. The estuary has exchanges with the Northern Gulf of America (NGA) through two

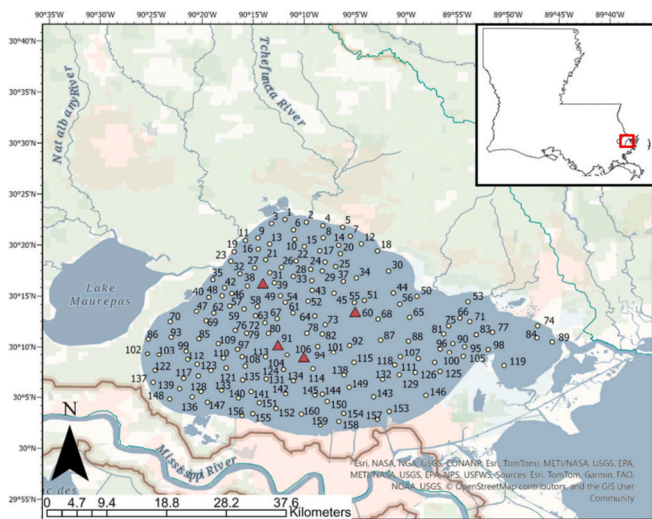


Fig. 1. Sampling map of the Lake Pontchartrain Estuary; white spots represent sampling locations for lake characterization, while red triangles indicate the four core sampling locations used for both diffusive and advective flux studies. The estuary is bisected by a causeway from its north to south shore, with sampling points ($n = 160$) distributed across the system.

southeastern outlets, Chef Menteur Pass and Rigolets, where tidal flushing results in an estuarine residence time of ~ 60 days (Swenson, 1980). In addition to the NGA, the estuary receives $\sim 6 \text{ km}^3$ of freshwater per year from tributaries that flow in from the northern reaches of its watershed (Wu and Xu, 2007). Most recent estimates of nutrient loading from the northern tributaries suggest an annual loading of 980 Mt yr^{-1} of ammonia-N, 980 Mt yr^{-1} of NO_x , 655 Mt yr^{-1} of TP, and 51 Tg yr^{-1} of sediment (Roblin, 2008). Episodically, a substantial source of freshwater and nutrients is delivered from the Mississippi River through the opening of the Bonnet Carré Spillway (White et al., 2019). During the high flood stage of the Mississippi River ($>35,000 \text{ m}^3 \text{ s}^{-1}$), the spillway can be opened to divert up to 20 % of the flood stage flow into the Lake Pontchartrain Estuary, significantly altering the salinity and nutrient dynamics within the estuary (White et al., 2009; Roy et al., 2012). Estuary salinity ranges between 2 and 9 during non-spillway years, with the potential to drop to ~ 0 during the opening of the Bonnet Carré Spillway (McCorquodale et al., 2009). The estuary has experienced an increasing occurrence of HABs, attributed to increased nutrient loading over the past several decades and increased spillway openings (Roy et al., 2013; Bargu et al., 2023).

2.2. Sediment sampling and analysis

A stratified random sampling scheme of 160 locations was determined for two subgroups across the estuary. The first section includes 100 sites that were randomly selected in the western portion of the estuary. The second section, including 60 randomly selected sites, was located on the eastern half, separated by the north-south Causeway, which bisects the estuary. This decision was based on previous studies by Roy et al. (2013) and DeLaune et al. (2004), which exhibited elevated spatial heterogeneity and nutrient concentration surrounding the northern tributaries and the Bonnet Carré Spillway to the west. The minimum sampling point lag distance was 1.5 km. Triplicate cores were collected at six locations during the initial mapping to document variability within stations.

Intact cores were collected from all 160 sites over five sampling trips to map the sediment distribution and composition across the estuary between November 2023 and February 2024. Samples of the top 10 cm of sediment were collected from the 160 sites using a modified piston core and sectioned in the field into 0–5 and 5–10 cm (320 samples)

increments. Sediment samples were placed in Ziploc bags, stored on ice, and returned to the laboratory for characterization. In the lab, the sections were weighed for total wet weight. A subset of the homogenized sample was weighed and dried at 60°C for 48 h or until a constant weight was achieved for moisture content calculations. Sediment bulk density was then calculated on a dry weight basis by dividing the total dry weight of the sample by the sample core volume. The dried sample was then ground using a mortar and pestle until fine-grained and weighed in a 50 mL glass beaker. The sample was burned at 550°C for 5 h and reweighed to determine the percent organic matter by loss-on-ignition (LOI) (White and Reddy, 1999). Following the method of Andersen (1976), the ashed sample was treated with 6 M HCl and heated on a hot plate until dry for total phosphorus (TP) analysis. The ash was then dissolved in 2.25 mL of 6 M HCl and diluted to 50 mL with DI water, and analyzed for TP using EPA method 365.1 (USEPA, 1993) on a SEAL AQ300 Automated Discrete Analyzer (SEAL Analytical, West Sussex, England). Total carbon (TC) and total nitrogen (TN) were determined from a subsample of the ground sediment and analyzed on a Costech Analytical Technologies Elemental Combustion System (Berkowitz et al., 2018). Metal determinations for Fe, Mg, Mn, Al, and Ca were performed on the digested TP sample using a Varian MPX ICP-OES. Total inorganic P (TIP) was extracted by treating $\sim 0.5 \text{ g}$ of dried, ground samples with 20 mL of 1 M HCl and shaking them longitudinally for 3 h (Reddy et al., 1998; Reddy et al., 2022). The samples were then centrifuged at 4000g and vacuum filtered using a $0.45 \mu\text{m}$ Whatman filter. Samples were analyzed colorimetrically for total phosphorus (TP) using EPA method 365.1 (USEPA, 1993) on a SEAL AQ300 Automated Discrete Analyzer (SEAL Analytical, West Sussex, England) with an SRP detection limit of 0.001 mg L^{-1} .

2.3. Diffusive flux experimental design

Following spatial mapping of the estuary, four sites with varying sediment characteristics and nutrient concentrations were selected for further sampling of intact cores for laboratory incubations (Fig. 2). K-means cluster analysis was performed on the 160 sites using TP, bulk density, moisture content, and organic content to determine four variable sediment regions across the estuary, and sites 38, 60, 91, and 106 were randomly selected from four clusters for incubation studies. These sites were assumed to be representative of their respective clusters' overall area. Triplicate intact 8 cm diameter cores were collected from the four sites within the estuary with a piston corer in May 2024, with care to maintain the sediment-water interface. Cores were sealed in the field with a rubber stopper and transported back to the laboratory, where they were placed in a temperature-controlled water bath at 25°C . The water column of the cores was then removed using a peristaltic pump and refilled using a drip fill method with filtered water to prevent disturbance of the uppermost sediment layer. The cores were continuously aerated twice a day for three-hour intervals with room air for 2 weeks during incubation and kept in the dark. Intact cores were maintained under aerobic conditions to represent the background conditions of the estuary, which is consistently aerobic, averaging $8 \pm 1.6 \text{ mg L}^{-1}$ rarely entering hypoxic conditions (LADEQ WQ1958001 site 0138). Seven mL water samples were collected using a syringe, pulled from 5 cm below the top of the 10 cm water column, daily for the first seven days of incubation, and then again on the fourteenth day. Samples were syringe filtered using a $0.45 \mu\text{m}$ Whatman disposable filter. Samples were acidified with concentrated H_2SO_4 to a pH below 2, as required for colorimetric analysis, and refrigerated until analysis. Following sampling, the column was refilled with seven mL of water to maintain a constant water column depth. Triplicate cores were also collected from the four sites for further sediment profiling. Cores were sectioned at 1-cm intervals for a higher resolution breakdown of physicochemical properties in the sediment layer across the selected sites.

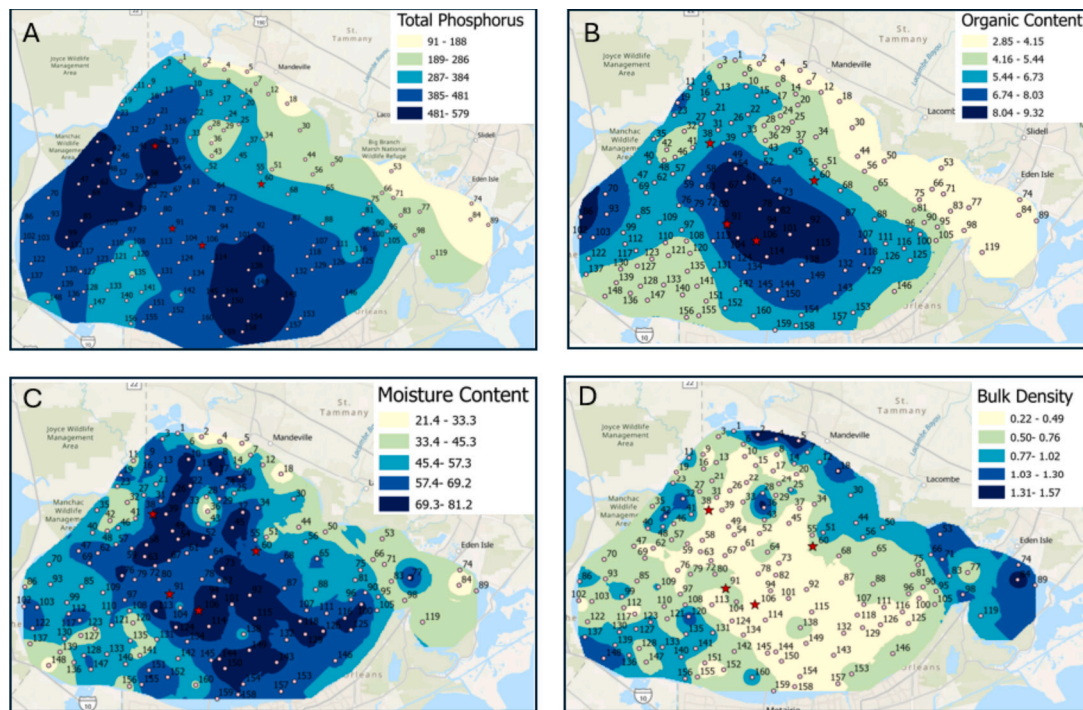


Fig. 2. Spatial distribution of sediment characteristics in the Lake Pontchartrain Estuary for the top 0–5 cm. A) Total phosphorus mg kg^{-1} , B) organic matter content (wt%), C) moisture content (%), D) bulk density (g cm^{-3}). Data are interpolated from the 160 sediment samples collected across the estuary through ordinary kriging

2.4. Advective flux experimental design

At each site from the diffusive flux experiment, four replicate cores were collected using 10 cm cores employing previously outlined methods in January 2025. Cores were transported back to the lab and stored at 4 °C. The advection experiment was run within one week of core collection to minimize the effects of sediment storage. Cores were outfitted on a gust erosion microcosm system (GEMs) to emulate variable shear stress on the sediment bed. The GEMs system applies a uniform shear stress ranging from 0.01 Pa to 0.45 Pa to the sediment bed layer by spinning magnetic disks at the top of a 10 cm water column. The 0.45 Pa is the highest level of applicable shear stress that falls within the system calibration (Green Eyes LLC, Easton, MD). The system operates by pumping water through a closed loop from a two-liter beaker reservoir, via precleaned tubes, into the erosional head, where it mixes with eroded sediment and core water, and then back into the reservoir. The system runs at 0.01 Pa for 20 min to reach equilibrium conditions without disturbing the sediment layer. Once the experiment began, shear stress was increased 5 times from 0.01 to 0.1, 0.2, 0.3, 0.4, and 0.45 every 20 min. Samples were collected at shear stress level three times, at the two-minute mark, the 10-minute mark, and the 19-minute mark, before the shear stress was increased. The 20-minute time interval was selected due to previous findings with the GEMs that found it to erode the sediment layer at each shear stress interval sufficiently (Li et al., 2022; Kleeberg et al., 2007). Water samples were collected from the well-mixed reservoir, approximately 5 cm below the surface. Following all sample collections, the reservoir was refilled with 7 mL of filtered water. All water samples were syringe-filtered through a 0.45 μm disposable filter and acidified using concentrated H_2SO_4 . Estimated flux rates were determined by a fitted linear regression across the three sample points at each shear stress level. Flux rates were determined by fitting a linear regression to the three points for each core at each shear stress level. These fluxes were then scaled up to per-day estimates for comparison across diffusive and advective flux.

2.5. Estuary field water quality data

Surface water quality data were provided from the Louisiana Department of Environmental Quality statewide water quality monitoring network (Site 0138). This station has long-term, monthly nutrient data collected from the Lake Pontchartrain Estuary Causeway in the estuary center for over the past 50 years. The database contains nitrate and total phosphorus (TP) data, among other water quality parameters. Additional water samples were collected during field sampling at the four stations used for incubation. Samples were collected in a one-L, pre-rinsed Nalgene bottle approximately 30 cm below the water surface. Samples were stored on ice and brought back to the lab to be filtered through 0.45 μm filters for SRP concentrations. A subset of the sample, both filtered and unfiltered, was autoclaved for the digestion of samples. These autoclaved samples were then run for TDP and TP.

2.6. Statistical analysis

All statistical analyses were performed in R version 4.2 and run in RStudio (Boston, MA, USA). Spatial mapping and Geostatistical analyses were performed in ArcGIS Pro. Data were tested for normality in R using a Shapiro-Wilk test. Following normality tests, spatial mapping was performed in ArcGIS Pro, where variograms were fitted for varying sediment characteristics. Following model selection, kriging was implemented to predict sediment dynamics across the estuary. Correlations within sediment characteristics were determined using Pearson correlation matrices. For the diffusive flux, due to the two zero flux rates, Mann-Whitney nonparametric tests were used to assess differences between sites. Analysis of variance was performed across sites for advective flux incubation results, with p -values <0.05 considered significant.

3. Results

3.1. Spatial mapping of estuary sediments

Sediment characteristics varied spatially across the 160 sites sampled in this study across Lake Pontchartrain Estuary. Generally, lower concentrations of metals and nutrients were measured to the east, while concentrations up to five times greater were observed in the estuary center and west. Little variability was observed across all measured characteristics between the sediment at depths of 0–5 cm and 5–10 cm. Subsequent results are for the top 5 cm of sediment in the estuary. Overall, the sediment was primarily inorganic, with an average wt% organic matter content of $5.95 \pm 2.51 \%$, and the largest concentrations of organic matter were found in the estuary center, ranging from 6.81 % to 9.34 %. Sediment bulk density was lowest in the estuary center ($0.22\text{--}0.49 \text{ g cm}^{-3}$) and highest in the east and southwest portions of the estuary, increasing to $1.02\text{--}1.57 \text{ g cm}^{-3}$. High bulk density was also found near the tributary mouths and the Bonnet Carré Spillway outlet (Fig. 2). Inorganic P (345 ± 123) was the dominant pool of TP (385 ± 140) at 90.0 % across all sites, with localized regions of organic P near the south around New Orleans (Fig. 2A). Concentrations of TP tended to be greater towards the west of the estuary, around the center. Lower concentrations were observed in the east near the mouth of the estuary. Metal content of the sediment was normally distributed for Fe, Mg, Mn, Al, and Ca. Calcium content was greatest near the estuary's inlet with the Gulf ($231,000 \text{ mg kg}^{-1}$) and lowest towards the western, freshwater reaches of the estuary (85.2 mg kg^{-1}). All other metals tended to be greatest in the center of the estuary, with Fe and Al being the most prevalent at ($26,900$ and $10,500 \text{ mg kg}^{-1}$) and ($25,800 \pm 12,100 \text{ mg kg}^{-1}$), respectively.

3.2. Water quality

Long-term water column TP at the single monitored estuary center station has nearly doubled over the past 40 years, increasing from a mean of 0.05 mg L^{-1} from 1980 to 2000 to an average of 0.1 mg L^{-1} from 2000 to 2024. Water column TP was higher on average in the cooler months of October–April, and lower in the warmer months of May–September (Fig. 3) The bioavailable P at the four sampling stations associated with this study varied with proximity to the north shore, with site 38 having the highest SRP content and decreasing at sites 60, 91,

Table 1

Water column phosphorus fractions across the four incubation sites. Mean concentrations (mg L^{-1}) are reported for soluble reactive phosphorus (SRP), total dissolved phosphorus (TDP), and total phosphorus (TP). Ratios of SRP to TDP and SRP to TP are also included to indicate the relative proportion of bioavailable phosphorus.

Site	SRP mg L^{-1}	TDP mg L^{-1}	TP mg L^{-1}	SRP/TDP mg L^{-1}	SRP/TP mg L^{-1}
38	0.041	0.036	0.068	1.139	0.603
60	0.008	0.011	0.022	0.727	0.364
91	0.007	0.009	0.013	0.778	0.538
106	0.009	0.010	0.017	0.900	0.529

and 106 near the estuary center. A majority of the TP in the water column was in the dissolved inorganic form at all sites, averaging 57 % SRP, with the remainder as particulate P, with only 5 % as dissolved organic P (Table 1).

3.3. Incubation site sediment characteristics

Across the four incubation sites, sediment characteristics were nearly homogeneous at sediment depths of 0–5 and 5–10 cm (Table 2). Based on a paired *t*-test, the only significant difference in P was observed at site 38, where the top 5 cm had significantly lower TP than the deeper sediment. At site 91, the top 5 cm of sediment had lower bulk density and higher moisture content. An ANOVA across sites revealed that TIP was significantly lower at site 60 than at the other three sites ($P < 0.05$). This difference was also observed for TP, with site 60 significantly lower than sites 106 and 91. A 1-cm increment analysis was performed for all four sites, and the highest P content was observed in the top 2 cm for all sites. In sites with a low average TP, a steep decrease was observed at a depth of 10 cm, where it was lowest. In sites with high overall TP, concentration showed little depth-related variability. This pattern is also inversely related to moisture content and bulk density, which were highest and lowest, respectively, in the top 2 cm of sediment. Bulk density was roughly 50 % lower in the top 2 cm vs the bottom 2 cm across all sites, with site 60 being the lowest overall in the top 1 cm. The top cm of sediment from site 60 had the second-highest TP, while it was initially registered as the lowest at 0–5 increments (Fig. 4 - TP & Table 2).

3.4. Diffusive flux

Across the four stations, the SRP concentration in the intact cores averaged $0.008 \pm 0.008 \text{ mg P L}^{-1}$ at the end of the two-week incubation. SRP release rates from sites 106 and 38 were below detection (Fig. 5). The water column for Site 106 was consistently measured at 0.01 mg P L^{-1} across the entire incubation following the initial water column refill. Site 91 had the highest post-incubation water column concentration of 0.017 mg L^{-1} over the two-week incubation period. Based on these diffusion rates, sediments from site 91 could release approximately $0.19 \text{ mg P m}^{-2} \text{ d}^{-1}$. Site 60 had roughly one-third the flux rate at $0.07 \text{ mg P m}^{-2} \text{ d}^{-1}$. Applying these flux rates across the estuary, upwards of $75.9 \text{ Mt SRP yr}^{-1}$ could be diffused annually as an estimate. This value is based on an average diffusive flux rate of $0.13 \text{ mg P m}^{-2} \text{ d}^{-1}$ from the two diffusive fluxes above the detection limit, applied across the estuaries' 1600 km^2 surface area.

3.5. Advective flux

Across all four sampling sites, the flux of SRP into the water column demonstrated a significantly positive relationship with increasing shear stress (Fig. 7). Over the two-hour incubation, water column SRP concentration increased steadily from an average of $0.006 \pm 0.003 \text{ mg SRP L}^{-1}$ across all sites to a maximum water column concentration of 0.023

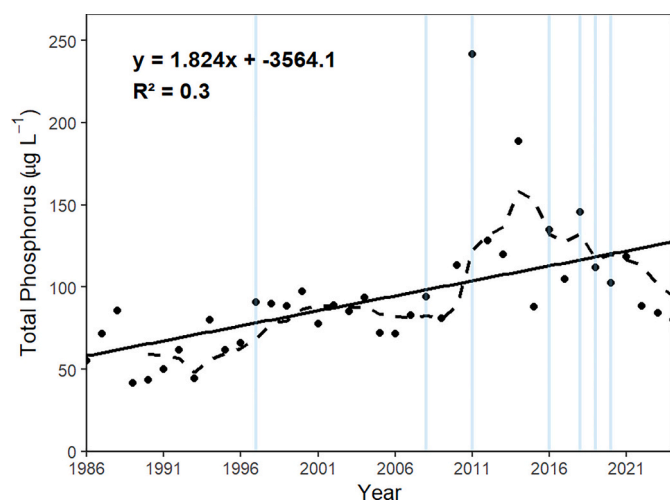


Fig. 3. Long-term total phosphorus (TP) trends in the water column of Lake Pontchartrain Estuary. Annual TP concentrations from 1986 to 2024 at LADEQ Site 138 (estuary center). Blue-shaded bars indicate the timing of Bonnet Carré Spillway openings, generally correlated with high TP values.

Source: <https://deq.louisiana.gov/page/ambient-water-quality-monitoring-data>.

Table 2

Sediment characteristics from four intact core incubation sites in Lake Pontchartrain Estuary. Mean $\pm$ standard deviation values are presented for two depth intervals (0–5 cm and 5–10 cm). Parameters include moisture content (%), bulk density (g cm^{-3}), organic matter content (wt%), total phosphorus (TP), inorganic phosphorus (TIP), total nitrogen (%), and total carbon (%).

Site	Depth cm	Moisture content %	Bulk density g cm^{-3}	Organic content wt%	Inorganic phosphorus mg kg^{-1}	Total phosphorus mg kg^{-1}	Total nitrogen %	Total carbon %
106	0–5	75.5 ± 4.3	0.57 ± 0.13	9.6 ± 1.7	400 ± 23	561 ± 68	0.19 ± 0.03	1.7 ± 0.35
	5–10	67.5 ± 8.4	0.81 ± 0.32	10.0 ± 7.2	404 ± 72	527 ± 41	0.16 ± 0.02	1.7 ± 0.48
38	0–5	71.1 ± 5.1	0.72 ± 0.24	9.4 ± 2.5	375 ± 61	488 ± 86	0.19 ± 0.04	1.9 ± 0.81
	5–10	64.1 ± 2.7	0.89 ± 0.16	10.0 ± 1.6	428 ± 59	463 ± 134	0.19 ± 0.10	1.6 ± 0.62
60	0–5	67.2 ± 14.6	0.89 ± 0.52	9.6 ± 4.1	321 ± 98	462 ± 136	0.2 ± 0.22	1.9 ± 0.83
	5–10	66.7 ± 2.3	0.87 ± 0.12	8.3 ± 3.0	318 ± 85	399 ± 40	0.12 ± 0.02	1.3 ± 0.25
91	0–5	75.7 ± 4.7	0.64 ± 0.12	7.5 ± 2.6	422.3 ± 46.8	549 ± 36	0.15 ± 0.03	2.0 ± 1.10
	5–10	69.6 ± 6.1	0.78 ± 0.1	6.9 ± 1.4	429.7 ± 29.4	532 ± 17	0.13 ± 0.01	1.4 ± 0.20

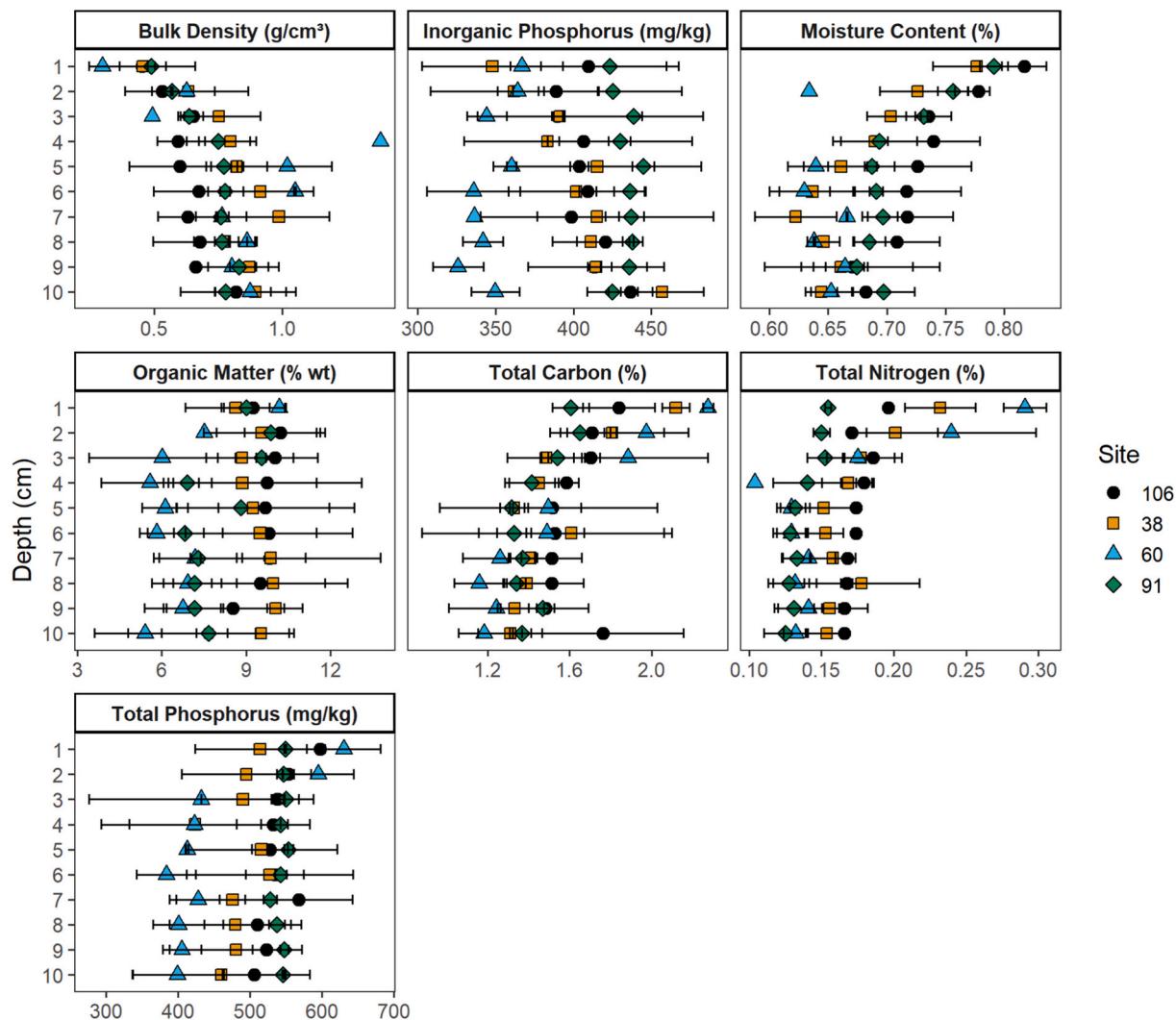


Fig. 4. Vertical profiles of sediment characteristics across incubation sites. Mean $\pm$ standard deviation of sediment moisture content, bulk density, organic matter content, and total phosphorus for each 1-cm depth increment (0–10 cm) at Sites 38, 60, 91, and 106.

$\pm 0.008 \text{ mg SRP L}^{-1}$ across the four sites. The release rate of SRP was approximately linear across all sites as the applied shear stress was increased, with the range of flux rates at a given shear stress ranging from 7.06 to $73.9 \text{ mg SRP m}^{-2} \text{ d}^{-1}$ observed for individual cores. The steepest flux rate was observed at sites 60 and 91. Slightly lower release rates, although not significantly so, were observed at 38 and 106, which fluxed roughly half the SRP on average during the experiment (Fig. 6). At all sites, SRP flux was lowest at a shear stress of 0.01 Pa . The highest

release rates were observed at a shear stress of 0.40 Pa . Based on an ANOVA, site 91 released significantly more SRP across each shear stress increment at an average rate of $38.3 \pm 15.4 \text{ mg SRP m}^{-2} \text{ d}^{-1}$. Site 38 released significantly less, with the lowest overall average flux rate of $22.3 \pm 7.15 \text{ mg SRP m}^{-2} \text{ d}^{-1}$. The release rate of SRP remained relatively linear at site 91 over each increase in shear stress. The remaining three sites all demonstrated significantly greater SRP release at 0.40 Pa shear stress, with release rates three times those at other shear stress

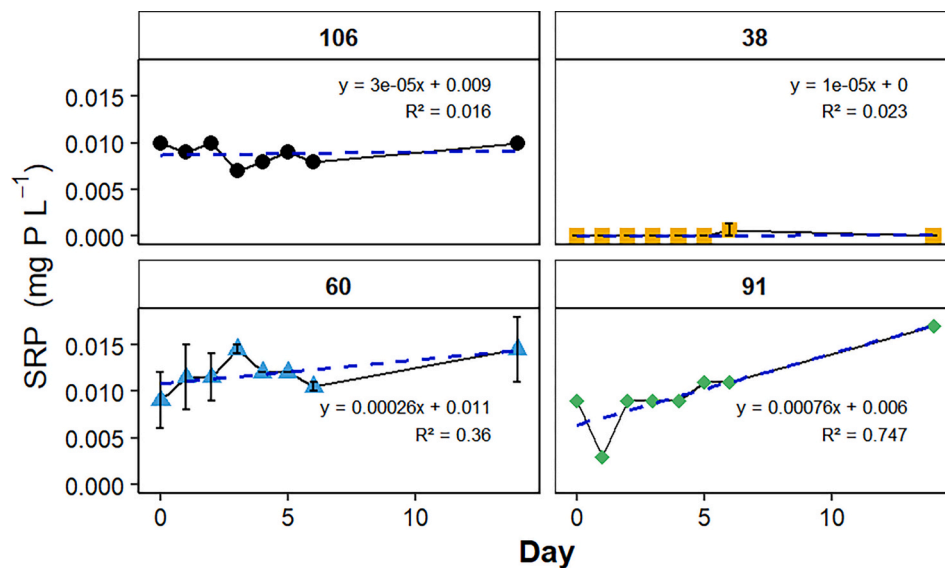


Fig. 5. Diffusive flux of soluble reactive phosphorus (SRP) from intact sediment cores. SRP flux ($\text{mg P m}^{-2} \text{d}^{-1}$) was calculated over a 14-day incubation at 25 °C under aerobic conditions. Data represent triplicate means $\pm$ standard deviation for each site.

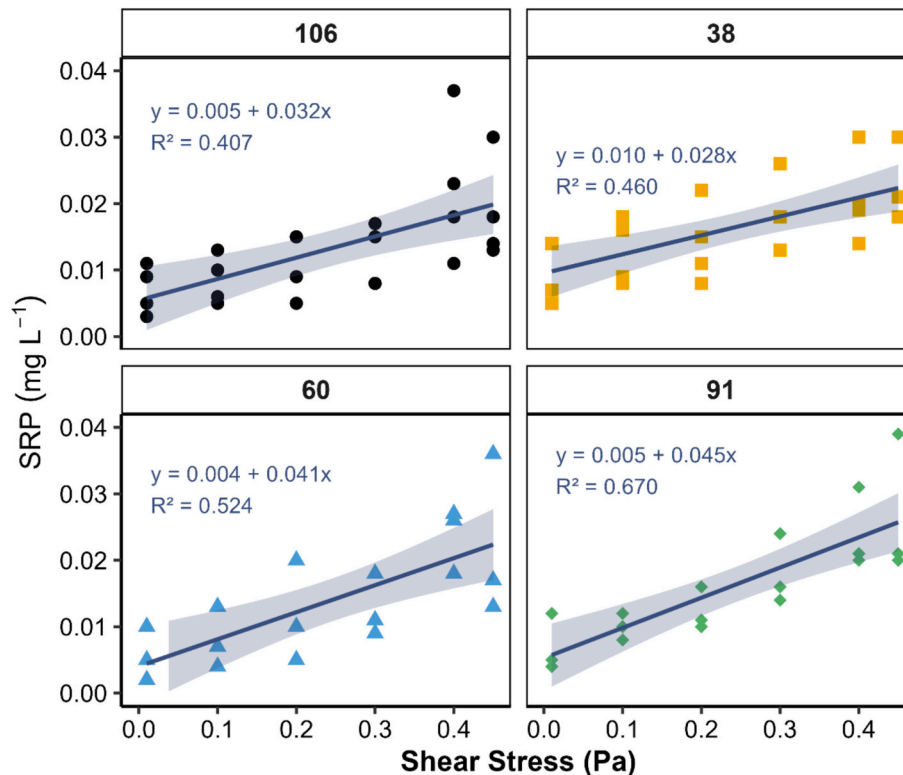


Fig. 6. Advective SRP flux at each shear stress level across incubation sites. SRP flux ($\text{mg P m}^{-2} \text{d}^{-1}$) estimated from water samples collected 19 min after each applied shear stress level (0.01–0.45 Pa). Data represent mean $\pm$ SD across replicates.

steps. Sites 91 and 60 had the highest flux rates per area at 25.9 ± 27.1 and $22.3 \pm 7.2 \text{ mg SRP m}^{-2} \text{d}^{-1}$ respectively. The highest flux rate at a given shear stress was observed at site 106 at 0.40 Pa, where rates were as high as $73 \text{ mg SRP m}^{-2} \text{d}^{-1}$. Across the four sites sampled, the average advective flux rate across all shear stress intervals was $30.8 \pm 9.2 \text{ mg SRP m}^{-2} \text{d}^{-1}$.

4. Discussion

4.1. Internal loading of SRP from estuary sediments

Following two weeks of steady-state incubations, SRP mobilization from the sediment to the water column was considerably lower than previous findings (Roy et al., 2012). Past studies have established a diffusive flux of SRP of roughly 1.8 to $3.2 \text{ mg P m}^{-2} \text{d}^{-1}$ under aerobic conditions equivalent to between 3 and 5 Mt of SRP across the estuary's

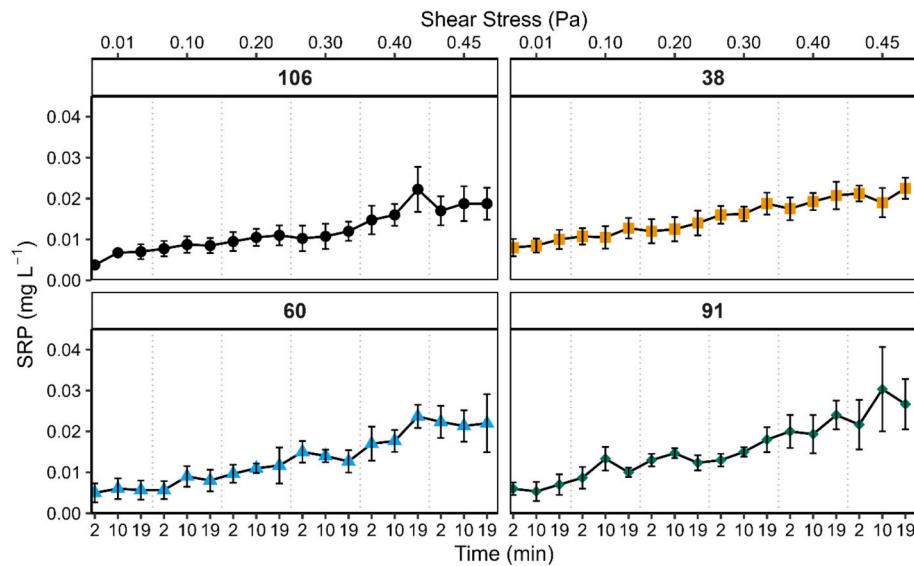


Fig. 7. SRP release over time from sediment as a function of increasing shear stress. Pooled data across all sites showing cumulative SRP concentrations in the water column (mg L^{-1}) in response to increasing shear stress during the 2-hour Gust Erosion Microcosm (GEMs) experiment.

1600 km^2 surface area daily (White et al., 2009; Bargu et al., 2011). Meanwhile, across all our sample cores, the average release rate was $0.13 \text{ mg P m}^{-2} \text{ d}^{-1}$ (0.21 Mt per day) from similar regions of the estuary, near the northern shore, and in the area of Bonnet Carré influence. This significant decline in the SRP diffused from the sediment is likely related to the fact that the spillway had not opened in the three years preceding our sample collection. This finding suggests that the impacts of P loading from the Mississippi River to the estuary may be of shorter duration than previously hypothesized. Previous works estimated that a substantial flux of sediment TP and SRP added by the operation of the spillway would take roughly six years to return to baseline under diffusive flux alone (Roy et al., 2012, 2017). Our results indicate that just three years after the last spillway opening, flux rates are very low, suggesting a more rapid return to pre-diversion baseline water quality conditions. This may be due to a changing form of sediment-bound P, as TP remains roughly the same concentration as reported in (Roy et al., 2012). While TP has continued to increase in the water column over this period (Fig. 4), the increase may be due to the higher spring discharge from northern tributaries or from alternative internal P loading pathways (Wu and Xu, 2007).

These findings suggest that advection, in concert with diffusion, plays a substantial role in regulating the internal loading of bioavailable P from the sediment into the overlying water column. In shallow systems like Lake Pontchartrain Estuary, regular mixing of the water columns due to surface wind and currents tends to keep the water column well-oxygenated, with anaerobic conditions at the surface of the sediment bed rarely observed (McCorquodale et al., 2002; White et al., 2009). Reduced mixing during low-wind, high-temperature summer months has been shown to intermittently limit the oxygen transported to the bottom waters, creating episodic anaerobic conditions in the southern part of the estuary (Haralampides et al., 2000; Poirrier, 2013). These episodic events may result in elevated P release rates as Iron bound P is reduced near the sediment water interface and fluxed into the water column.

Weather patterns over the estuary are typically constrained to two seasons, a cooler autumn and winter from October to April, and warmer summer months with less prevalent winds occurring from May to September (Mortimer et al., 1988). Winter weather systems are often characterized by frequent cold fronts and storms (Georgiou and McCorquodale, 2002; Hiatt et al., 2019). Summer is characterized by warm, calm days, aside from periodic hurricanes and tropical depressions that can result in episodic, highly dynamic changes to the

systems through winds exceeding 33 m s^{-1} (Stone et al., 1997; Van Metre et al., 2006). There was a measured 20–30 % increase in TP, attributed to Fe–P in the sediments of the western portion of the estuary after a one-month opening of the Bonnet Carré Spillway in 2011 (Roy et al., 2017). This study documented a large amount of introduced sediment and P that had accumulated in a transitory (Fe-bound), which would increase the accumulation of SRP in sediment porewater over time under anaerobic conditions, thereby increasing diffusion rates. The increased concentration of TP in the top sediment layer is likely due to diffusive SRP binding to Fe as it approaches the aerobic surface layer. The low diffusion rates observed across our sites may be explained by the highly aerobic conditions at the upper sediment layer, as the oxidized Fe at the surface acts as a cap on low rates of SRP diffusion across the interface (Immers et al., 2015). Three years after the last spillway opening, the low or absence of bioavailable diffusive P flux in our study may support an additional process from storm-induced mixing, which limited the diffusion rate of SRP.

4.2. Implications of storm events on SRP loading

Findings from our advective flux experiment showed substantially greater episodic SRP flux than diffusion, highlighting the significance of wind-driven sediment perturbations. These events, on average could result in a flux of ~ 240 times more SRP at approximately 50 Mt d^{-1} when generalized across the estuary's surface area, when compared to diffusive flux (0.21 Mt d^{-1}). Although advective events are episodic and diffusion is a constant phenomenon, the significantly higher flux rates indicate that these events could cause short-term bursts of nutrients rather than the slower diffusion. The four sampling stations exhibited a consistent increase in SRP release to the water column in response to increases in shear stress. Unlike in previous studies that have utilized a GEMs system (Gust and Müller, 1997; Dickhudt et al., 2009; Hurst et al., 2019), suspension of sediment and P concentration from Lake Pontchartrain Estuary appeared to increase linearly with shear stress, rather than following a stepwise increase with shear stress increases. This pattern may suggest that the 20-min time interval typically employed is insufficient to completely erode sediment at each shear stress level in certain systems. This difference could be due to the low-organic, high-clay-content sediments typical of Lake Pontchartrain Estuary, which increase sediment cohesion (Delaune et al., 2008; Wright et al., 2022). As previously stated, flux rates from our study tended to increase linearly with shear stress until the 0.40 Pa interval. This is with the

exception of site 91, which had a linear release rate across the entire experiment and diffused the greatest amount of SRP. This may be due to its proximity to the Bonnet Carré Spillway, which changed the geochemistry of the sediments through a significant increase in the iron concentration in the sediment profile and fraction of Fe—P (Roy et al., 2017). At this setting, flux rates increased rapidly by roughly triple the rate in lower shear stress levels, indicating a potential threshold for maximum release rates. The lowest setting of shear stress applied (0.1 Pa) corresponds to about 6 m s^{-1} wind conditions that are slightly below the average for the estuary of 5 m s^{-1} (Li et al., 2018), indicating that wind has the potential to steadily allow for low levels of sediment erosion and a minimum P release of $17.9 \text{ mg P m}^{-2} \text{ d}^{-1}$ during winds $\geq 6 \text{ m s}^{-1}$. While strong wind conditions are discontinuous, this minimum advective flux remains ~ 140 times that of the average observed diffusive flux ($0.13 \text{ mg P m}^{-2} \text{ d}^{-1}$) and can reach up to ~ 240 times when considering the highest shear stress applied in this study. Based on modeling by the Army Corps of Engineers, the maximum sediment resuspension in Lake Pontchartrain Estuary occurs above of the range of our lab simulation, at ~ 1 Pascal of shear stress, indicating even greater release rates of SRP ($\sim 16.9 \text{ m s}^{-1}$ wind) (Stone and Deegan, 1980). This rate is far greater than the 48 x greater advective P flux when compared to diffusive flux found for the hypereutrophic Lake Apopka in FL although the shear stress was not calibrated in that study (Reddy et al., 1996).

The GEMs device has a maximum practical shear stress of 0.45 Pa ($\sim 9 \text{ m s}^{-1}$), which more closely aligns with the mean wind conditions observed in winter cold fronts (Sha et al., 2018; Hiatt et al., 2019). Cold fronts in southeastern coastal Louisiana occur on average every 3 to 7 days during the autumn and winter months and can last for up to 24 h, with residual wave disturbance as the front passes. These fronts potentially release as much as 49.3 Mt of SRP per event (Hiatt et al., 2019). This pattern supports that the bioavailable phosphorus (P) beneath the sediment-water interface is regularly released into the water column during the winter months following storms and more frequent wind events, which disturb the sediment layer (Hiatt et al., 2019; Stone and Deegan, 1980). The flux rate under conditions consistent with winter cold fronts (9 m s^{-1}) was roughly three times greater ($52.8 \text{ mg P m}^{-2} \text{ d}^{-1}$) than the lowest level wind-induced (6 m s^{-1}) shear stress, which released ($17.9 \text{ mg P m}^{-2} \text{ d}^{-1}$). Once resuspended, the resuspended SRP's fate and duration within the water column are currently not fully understood. The SRP can be reburied in the sediment as the storm passes, sorbing to settling particles of Fe/Al oxides or precipitating with Ca/Mg-based minerals. The SRP could move into the biotic compartment through phytoplankton uptake or could flush out into the Gulf following resuspension. This significant knowledge gap remains in our understanding of the estuary propensity to retain P, once loaded to the system. Greater retention would lead to consistent eutrophication over time.

Research has shown that the primary Fe-bound P in river sediment transitions to more stable Ca-bound P once exposed to seawater on the shelf (Adhikari et al., 2015) or in coastal bays (Potter et al., 2025). This pattern may suggest that as the resuspended SRP flows towards the gulf, it may become associated with Ca that is found in substantially higher quantities in seawater. High SRP loading in the winter months tends to coincide with lower uptake by primary producers due to the lower temperatures and elevated turbidity, which act as inhibitors (Sapkota et al., 2023). Enhanced flushing during the winter months may therefore suggest a mechanism contributing to the depleted sediment SRP stores observed in our diffusive experiment. As episodic wind events increase shear stress and SRP release, particularly in the winter months, less porewater SRP is present for passive diffusion. These results suggest seasonal differences in internal P loading capacity.

4.3. Limitations and implications of internal phosphorus loading on harmful algal blooms

Findings from this study demonstrate that the internal loading of P

within shallow waterbodies like the Lake Pontchartrain Estuary is more dynamic than previously understood. Previous studies surmised that regular river spillway openings would result in an increased diffusive release of SRP into the water column over six years due to a 30 % increase in sediment Fe—P (Roy et al., 2017). This release fueled HABs during the ensuing warmer summer months (Roy et al., 2012). Summer quiescent and warmer water column conditions are more likely to promote and facilitate the development of HABs (Ramos et al., 2023). Initial findings suggested diffusion to be the primary mechanism for summertime P release immediately following the opening of the Spillway (Roy et al., 2012). Our findings, however, suggest that diffusion becomes substantially less important as a driver of SRP release after 3 years post-spillway opening. Furthermore, wind and storm-based advective processes are a significant driver for sediment P release, particularly in discrete temporal intervals. Based on local weather patterns, winter cold fronts that bring repeated strong winds to the estuary are a potential pathway for the release of SRP throughout the winter months, coinciding with when the estuary is less biologically productive (Miller et al., 2007; Huang and Li, 2019; Hiatt et al., 2019). This pattern was seen after the 2016 spillway opening in the winter, which did not lead to a subsequent diatom algal bloom due to light limitations and colder temperatures (Roy et al., 2016; Sapkota et al., 2023). Wind-driven release of nutrients, therefore, may be a potential regulating mechanism that ultimately leads to less SRP available in the system during the ensuing summer months.

One exception is related to episodic summer events, including tropical depressions and hurricanes, which introduce high winds exceeding 33 m s^{-1} . Based on our findings, these events have the potential to release large quantities of SRP in short bursts. This phenomenon was observed in 2019 with Hurricane Barry, which resulted in a large resuspension event of SRP and subsequent HAB during the summer (Snow et al., 2024; Cangialosi et al., 2019). Subsequent studies have modeled and confirmed increased HAB occurrence in the days following wind events, driven by vertical nutrient mixing and phytoplankton distribution (Wang et al., 2016; Liao et al., 2021; Yin et al., 2004). As established in this study, consistent wind intensity can stir bottom sediments, releasing large amounts of inorganic nutrients into the water column. However, the disturbance of sediment can also increase water turbidity and decrease the light availability required for these primary producers, potentially serving as a feedback loop against an immediate bloom. Philips et al. (2020) found that HABs can proliferate when environmental conditions are favorable, as typically seen in the summer, and that nutrient levels remain elevated following a decrease in turbidity. This suggests that watershed managers account for storms as drivers of SRP release and potential triggers of HABs. Further studies are needed to validate this impact, although field-based evidence from numerous studies supports the proliferation of HABs following elevated nutrient levels from storms and subsequent turbidity drops (Hammond et al., 2025; Bargu et al., 2023; Roy et al., 2017). These findings suggest that in years with elevated winter storm activity, summer HABs would be depressed due to lower levels of diffusively available sediment P.

One limitation of this study is the lack of measured anaerobic diffusion flux rates. While the estuary does not routinely undergo regular anoxia conditions, episodic, localized anaerobic events could be a source of SRP release due to the reduction of Fe—P. A previous study noted that under an anaerobic water column, P flux was twice that of aerobic flux, where Fe-rich sediment is more common (Roy et al., 2012). This condition should be taken into account if attempting to model HAB occurrence in other estuarine systems that experience significant hypoxia.

Additionally, to prevent corrosive damage to the GEM's device, filtered, zero-salinity water was used. Salinity levels have implications for coagulation and subsequent flocculation of suspended sediment. Certain metals, such as Fe, are preferentially removed from the water column via flocculation with OM (Sholkovitz, 1976), while also co-precipitating with phosphate (Spiteri et al., 2008). Further studies

should investigate salinity levels, as well as the settling time of suspended sediment and the concomitant change in water column SRP, to better understand the fate of the released P.

5. Conclusions

This study determined the internal sediment loading of SRP to the water column through both diffusion and shear stress-induced advective flux. Diffusive flux is between 10 and 30 times lower ($0.13 \text{ mg P m}^{-2} \text{ d}^{-1}$) than previous estimates ($1.8\text{--}3.2 \text{ mg P m}^{-2} \text{ d}^{-1}$), suggesting that spillway openings have a shorter post-event impact on SRP release than predicted. Furthermore, advective flux resulted in up to ~ 240 times greater SRP release compared to diffusive flux. Advective flux ranged from 7.1 to $73.9 \text{ mg P m}^{-2} \text{ d}^{-1}$ at select shear stress intervals and sites, overall averaging $30.8 \text{ mg P m}^{-2} \text{ d}^{-1}$ across the experiment. This study demonstrates that, while diffusion is a continuous, slow process, the episodic, large-scale areal flux of SRP driven by wind-driven shear stress can account for the majority of internal loading throughout the winter months. This depletion would reduce SRP availability for diffusion under aerobic conditions in the summer months, when environmental conditions are more favorable to HAB proliferation. While this study focused on one estuary system in Louisiana, coastal systems worldwide are highly dynamic. Variability in tidal regimes, storm activity, and hydrologic input impacts localized sediment shear stress and phosphorus release. These networks should therefore account for advective fluxes in internal loading assumptions, given their pronounced role in P release over shorter time scales. These findings suggest that winter storms may act as a regulatory mechanism for the internal loading capacity of SRP from estuarine sediments.

CRedit authorship contribution statement

Lee Potter: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Conceptualization. **John R. White:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- Adhikari, P.L., White, J.R., Maiti, K., Nguyen, N., 2015. Phosphorus speciation and sedimentary phosphorus release from the Gulf of Mexico sediments: implication for hypoxia. *Estuar. Coast. Shelf Sci.* 164, 77–85.
- Andersen, J.M., 1976. An ignition method for determination of total phosphorus in lake sediments. *Water Res.* 10 (4), 329–331.
- Anderson, D.M., Burkholder, J.M., Cochlan, W.P., Glibert, P.M., Gobler, C.J., Heil, C.A., Kudela, R.M., Parsons, M.L., Rensel, J.J., Townsend, D.W., Trainer, V.L., Vargo, G.A., 2008. Harmful algal blooms and eutrophication: examining linkages from selected coastal regions of the United States. *Harmful Algae* 8 (1), 39–53.
- Bargu, S., White, J.R., Li, C., Czubakowski, J., Fulweiler, R.W., 2011. Effects of freshwater input on nutrient loading, phytoplankton biomass, and cyanotoxin production in an oligohaline estuarine lake. *Hydrobiologia* 661, 377–389.
- Anderson, D.M., Fensin, E., Gobler, C.J., Hoeglund, A.E., Hubbard, K.A., Kulis, D.M., Landsberg, J.H., Lefebvre, K.A., Provoost, P., Richlen, M.L., Smith, J.L., Solow, A.R., Trainer, V.L., 2021. Marine harmful algal blooms (HABs) in the United States: History, current status and future trends. *Harmful Algae* 102, 101975.
- Bargu, S., Hiatt, M., Maiti, K., Miller, P., White, J.R., 2023. The future of Cyanobacteria toxicity in estuaries undergoing pulsed nutrient inputs: a case study from coastal Louisiana. *Water* 15 (21), 3816.
- Berkowitz, J.F., VanZomerem, C.M., Piercy, C.D., White, J.R., 2018. Evaluation of coastal wetland soil properties in a degrading marsh. *Estuar. Coast. Shelf Sci.* 212, 311–317.
- Boydton, W.R., Ceballos, M.A.C., Bailey, E.M., Hodgkins, C.L.S., Humphrey, J.L., Testa, J. M., 2018. Oxygen and nutrient exchanges at the sediment-water interface: a global synthesis and critique of estuarine and coastal data. *Estuar. Coasts* 41 (2), 301–333.
- Cangialosi, J.P., Hagen, A.B., Berg, R., 2019. PDF. Hurricane Barry. National Hurricane Center Tropical Cyclone Report. https://www.nhc.noaa.gov/data/tcr/AL022019_Barry.pdf (31pp, last accessed 9/17/25).
- Chambers, P.A., Meissner, R., Wrona, F.J., Rupp, H., Guhr, H., Seeger, J., Culp, J.M., Brua, R.B., 2006. Changes in nutrient loading in an agricultural watershed and its effects on water quality and stream biota. *Hydrobiologia* 556, 399–415.
- Cullen, P., 1986. Managing nutrients in aquatic systems: the eutrophication problem. In: Decker, P., Williams, W.D. (Eds.), *Limnology in Australia*. Springer Netherlands, Dordrecht, pp. 539–554.
- Cymbala, J., Ogdahl, M., Steinman, A.D., 2008. Phytoplankton response to light and internal phosphorus loading from sediment release. *Freshw. Biol.* 53 (12), 2530–2542.
- DeLaune, R.D., Devai, I., Patrick Jr., W.H., 2004. Total mercury, methylmercury and other toxic heavy metals in Lake Pontchartrain Basin (USA). *J. Environ. Sci. Health A* 34 (13), 1657–1662.
- Delaune, R.D., Gambrell, R.P., Jugsujinda, A., Devai, I., Hou, A., 2008. Total Hg, methyl Hg and other toxic heavy metals in a northern Gulf of Mexico estuary: Louisiana Pontchartrain basin. *J. Environ. Sci. Health A* 43 (9), 1006–1015.
- Dickhudt, P.J., Friedrichs, C.T., Schaffner, L.C., Sanford, L.P., 2009. Spatial and temporal variation in cohesive sediment erodibility in the York River estuary, eastern USA: a biologically influenced equilibrium modified by seasonal deposition. *Mar. Geol.* 267 (3–4), 128–140.
- Georgiou, I., McCorquodale, J.A., 2002. Stratification and circulation in Lake Pontchartrain. In: Spaulding, M.L. (Ed.), *Estuarine and Coastal Modeling* (2001), Proceedings of the 7th International Conference. St. Petersburg FL, pp. 140–151.
- Gu, C., Joshi, S., Fischel, M.H., Tomaszewski, E.J., Northrup, P., Sparks, D.L., 2024. Saltwater intrusion increases phosphorus abundance and alters availability in coastal soils with implications for future sea level rise. *Sci. Total Environ.* 931, 172624.
- Gust, G., Müller, V., 1997. Interfacial hydrodynamics and entrainment functions of currently used erosion devices. In: Burt, N. (Ed.), *Cohesive Sediments: 4th Nearshore and Estuarine Cohesive Sediment Transport Conference, INTERCOH '94*, 11–15 July 1994: Wallingford, England, UK. Wiley, New York, USA, pp. 149–174.
- Hallegraeff, G.M., Anderson, D.M., Belin, C., Bottein, M.Y., Bresnan, E., Chinain, M., Enevoldsen, H., Iwataki, M., Karlson, B., McKenzie, C.H., Sunesen, I., Pitcher, G.C., Provoost, P., Richardson, A., Schweibold, L., Tester, P.A., Trainer, V.L., Yñiguez, A. T., Zingone, A., 2021. Perceived global increase in algal blooms is attributable to intensified monitoring and emerging bloom impacts. *Commun. Earth Environ.* 2, 117.
- Hallett, C.S., Valesini, F.J., Clarke, K.R., Hoeksema, S.D., 2016. Effects of a harmful algal bloom on the community ecology, movements and spatial distributions of fishes in a microtidal estuary. *Hydrobiologia* 763, 267–284.
- Hammond, C., Bargu, S., Turner, C., Hiatt, M., White, J.R., 2025. The impact of seasonal hydrological disturbances on the presence and formation of toxic cyanobacteria blooms on coastal Louisiana. *Front. Mar. Sci.* 12, 1618353.
- Haralampides, K., Georgiou, I.Y., McCorquodale, J.A., 2000. Water quality impacts on the Lake Pontchartrain estuarine system. In: *Proc. of the 75th Annual Conference of the Water Environ. Fed.*, pp. 15–21 (Last accessed 9/17/25: researchgate.net/publication/233499021/WATER_QUALITY_IMPACTS_ON_THE_LAKE_PONTCHARTRAIN_ESTUARINE_SYSTEM#fullTextFileContent).
- Heil, C.A., Muni-Morgan, A.L., 2021. Florida's harmful algal bloom (HAB) problem: escalating risks to human, environmental and economic health with climate change. *Front. Ecol. Evol.* 9, 646080.
- Hiatt, M., Snedden, G., Day, J.W., Rohli, R.V., Nyman, J.A., Lane, R., Sharp, L.A., 2019. Drivers and impacts of water level fluctuations in the Mississippi River delta: implications for delta restoration. *Estuar. Coast. Shelf Sci.* 224, 117–137.
- Huang, W., Li, C., 2019. Spatial variation of cold front wind-driven circulation and quasi-steady state balance in Lake Pontchartrain Estuary. *Estuar. Coast. Shelf Sci.* 224, 154–170.
- Hurst, N.R., White, J.R., Xu, K., Ren, M., 2019. Nitrate reduction rates in sediments experiencing turbulent flow conditions. *Ecol. Eng.* 128, 33–38.

- Immers, A.K., Bakker, E.S., Van Donk, E., Ter Heerdt, G.N.J., Geurts, J.J.M., Declerck, S. A.J., 2015. Fighting internal phosphorus loading: an evaluation of the large scale application of gradual Fe-addition to a shallow peat lake. *Ecol. Eng.* 83, 78–89.
- Jickells, T., 2005. External inputs as a contributor to eutrophication problems. *J. Sea Res.* 54 (1), 58–69.
- Kleeberg, A., Hupfer, M., Gust, G., 2007. Phosphorus entrainment due to resuspension in a lowland river, Spree, NE Germany—a laboratory microcosm study. *Water Air Soil Pollut.* 183 (1), 129–142.
- Li, C., Walker, N., Hou, A., Georgiou, I., Roberts, H., Laws, E., McCorquodale, J.A., Weeks, E., Li, X., Crochet, J., 2008. Circular plumes in Lake Pontchartrain estuary under wind straining. *Estuar. Coast. Shelf Sci.* 80 (1), 161–172.
- Li, C., Huang, W., Chen, C., Lin, H., 2018. Flow regimes and adjustment to wind-driven motions in Lake Pontchartrain estuary: a modeling experiment using FVCOM. *J. Geophys. Res. Oceans* 123 (11), 8460–8488.
- Li, Q., Gray, K.E., Jaisi, D.P., 2022. Relative roles of sediment transport and localized erosion on phosphorus load in the lower Susquehanna River and its mouth in the Chesapeake Bay, USA. *J. Geophys. Res. Biogeosci.* 127 (8), e2022JG006944.
- Liao, A., Han, D., Song, X., Yang, S., 2021. Impacts of storm events on chlorophyll-a variations and controlling factors for algal bloom in a river receiving reclaimed water. *J. Environ. Manag.* 297, 113376.
- McCorquodale, J.A., Georgiou, I., Haralampides, K., 2002. Bottom dissolved oxygen and salinity in Lake Pontchartrain. In: *Environmental Atlas of the Lake Pontchartrain Basin, Lake Pontchartrain Basin Foundation, US Geological Survey Open File Report, 02-206*. Last Accessed 9/17/25: <https://pubs.usgs.gov/of/2002/of02-206/env-issues/dissolved-oxygen.html>.
- McCorquodale, J.A., Roblin, R.J., Georgiou, I.Y., Haralampides, K.A., 2009. Salinity, nutrient, and sediment dynamics in the Pontchartrain Estuary. *J. Coast. Res.* 10054, 71–87.
- Miller, R.L., McKee, B.A., D'Sa, E.J., 2007. Monitoring bottom sediment resuspension and suspended sediments in shallow coastal waters. In: Miller, R.L., Del Castillo, C.E., McKee, B.A. (Eds.), *Remote Sensing of Coastal Aquatic Environments: Technologies, Techniques and Applications*. Springer Netherlands, Dordrecht, pp. 259–276.
- Mortimer, E.B., Johnson, G.A., Lau, H.W., 1988. Major arctic outbreaks affecting Louisiana. In: *The Second National Winter Weather Workshop Was Held 26-30 September 1988 in Raleigh, North Carolina*, p. 174. This volume contains refereed articles from the participants. Last accessed 9/17/25: https://repository.library.noaa.gov/view/noaa/7168/noaa_7168_DS1.pdf.
- Nixon, S.W., 1995. Coastal marine eutrophication: a definition, social causes, and future concerns. *Ophelia* 41 (1), 199–219.
- Philips, E.J., Badylak, S., Nelson, N.G., Havens, K.E., 2020. Hurricanes, El Niño and harmful algal blooms in two sub-tropical Florida estuaries: direct and indirect impacts. *Sci. Rep.* 10 (1), 1910.
- Pinckney, J.L., Paerl, H.W., Tester, P., Richardson, T.L., 2001. The role of nutrient loading and eutrophication in estuarine ecology. *Environ. Health Perspect.* 109 (Suppl. 5), 699–706.
- Poirrier, M.A., 2013. Effects of closure of the Mississippi River Gulf Outlet on saltwater intrusion and bottom water hypoxia in Lake Pontchartrain. *Gulf Caribb. Res.* 25 (1), 105–109.
- Potter, L., Mates, P., White, J.R., 2025. Implications of river reconnection on phosphorus cycling in coastal wetlands. *Sci. Total Environ.* 974, 179205.
- Ramos, I.M.S., Crooke, B., Seegers, B., Cetinić, I., Cambazoglu, M.K., Armstrong, B., 2023. Spatial and temporal characterization of cyanobacteria blooms in the Mississippi Sound and their relationship to the Bonnet Carré Spillway openings. *Harmful Algae* 127, 102472.
- Reddy, K.R., Fisher, M.M., Ivanoff, D., 1996. Resuspension and diffusive flux of nitrogen and phosphorus in a hypereutrophic lake. *J. Environ. Qual.* 25 (2), 363–371.
- Reddy, K.R., Wang, Y., DeBusk, W.F., Fisher, M.M., Newman, S., 1998. Forms of soil phosphorus in selected hydrologic units of the Florida Everglades. *Soil Sci. Soc. Am. J.* 62 (4), 1134–1147.
- Reddy, K.R., Newman, S., Osborne, T.Z., White, J.R., Fitz, C., 2011. Phosphorus cycling in the Greater Everglades ecosystem: legacy phosphorus implications for management and restoration. *Crit. Rev. Environ. Sci. Technol.* 41 (6), 149–186.
- Reddy, K.R., DeLaune, R.D., Inglett, P.W., 2022. *Biogeochemistry of Wetlands: Science and Applications*. CRC press, Boca Raton, FL.
- Roblin, R.J., 2008. *Water Quality Modeling of Freshwater Diversions in the Pontchartrain Estuary*. University of New Orleans, New Orleans, Louisiana (Master's thesis, 220p).
- Roth, D., 1998. A historical study of tropical storms and hurricanes that have affected Southwest Louisiana and Southeast Texas. NOAA, Technical Attachment SR/SSD, 98-16. Last accessed 9/17/25: <https://www.weather.gov/lch/lchhur>.
- Roy, E.D., Nguyen, N.T., Bargu, S., White, J.R., 2012. Internal loading of phosphorus from sediments of Lake Pontchartrain (Louisiana, USA) with implications for eutrophication. *Hydrobiologia* 684 (1), 69–82.
- Roy, E.D., White, J.R., Smith, E.A., Bargu, S., Li, C., 2013. Estuarine ecosystem response to three large-scale Mississippi River flood diversion events. *Sci. Total Environ.* 458, 374–387.
- Roy, E.D., Smith, E.A., Bargu, S., White, J.R., 2016. Will Mississippi River diversions designed for coastal restoration cause harmful algal blooms? *Ecol. Eng.* 91, 350–364.
- Roy, E.D., Nguyen, N.T., White, J.R., 2017. Changes in estuarine sediment phosphorus fractions during a large-scale Mississippi River diversion. *Sci. Total Environ.* 609, 1248–1257.
- Sapkota, Y., Bargu, S., White, J.R., 2023. Temporally-displaced Mississippi River spring flood pulse shows muted aquatic ecosystem response in estuarine waters: a climate change warning for coastal foodwebs. *Sci. Total Environ.* 874, 162623.
- Sha, X., Xu, K., Bentley, S.J., Robichaux, P.A., 2018. Characterization and modeling of sediment settling, consolidation, and suspension to optimize coastal Louisiana restoration. *Estuar. Coast. Shelf Sci.* 203, 137–147.
- Sholkovitz, E.R., 1976. Flocculation of dissolved organic and inorganic matter during the mixing of river water and seawater. *Geochim. Cosmochim. Acta* 40 (7), 831–845.
- Smith, V.H., 2003. Eutrophication of freshwater and coastal marine ecosystems a global problem. *Environ. Sci. Pollut. Res.* 10, 126–139.
- Snow, C., Bargu, S., Hammond, C.N., Hiatt, M., White, J.R., 2024. Effect of Mississippi River discharge plume on temporal and spatial variability of toxic cyanobacteria in an oligohaline estuary. *Hydrobiologia* 851 (1), 87–103.
- Søndergaard, M., Jensen, J.P., Jeppesen, E., 2003. Role of sediment and internal loading of phosphorus in shallow lakes. *Hydrobiologia* 506, 135–145.
- Spieteri, C., Van Cappellen, P., Regnier, P., 2008. Surface complexation effects on phosphate adsorption to ferric iron oxyhydroxides along pH and salinity gradients in estuaries and coastal aquifers. *Geochim. Cosmochim. Acta* 72 (14), 3431–3445.
- Stone, I.H., Deegan, L.A., 1980. Preliminary modeling of the Lake Pontchartrain ecosystem by computer simulation. In: *Environmental Analysis of Lake Pontchartrain, Louisiana, Its Surrounding Wetlands, and Selected Land Uses, US Army Corp of Engineers Report*, pp. 1–19. Last accessed 9/17/25: https://repository.library.noaa.gov/view/noaa/42179/noaa_42179_DS1.pdf.
- Stone, G.W., Grymes III, J.M., Dingler, J.R., Pepper, D.A., 1997. Overview and significance of hurricanes on the Louisiana coast, USA. *J. Coast. Res.* 656–669.
- Swenson, E.M., 1980. General hydrography of Lake Pontchartrain, Louisiana. In: *Environmental Analysis of Lake Pontchartrain, Louisiana, Its Surrounding Wetlands, and Selected Land Uses US Army Corp of Engineers Report*, pp. 57–155. Last accessed 9/17/25: https://repository.library.noaa.gov/view/noaa/42179/noaa_42179_DS1.pdf.
- Turner, R.E., 2024. Water quality at the end of the Mississippi River for 120 years: the agricultural imperative. *Hydrobiologia* 851 (5), 1219–1239.
- Turner, R.E., Rabalais, N.N., 1994. Coastal eutrophication near the Mississippi river delta. *Nature* 368 (6472), 619–621.
- Turner, R.E., Dortch, Q., Justic, D., Swenson, E.M., 2002. Nitrogen loading into an urban estuary: Lake Pontchartrain (Louisiana, USA). *Hydrobiologia* 487, 137–152.
- U.S. EPA, 1993. Methods for Determination of Inorganic Substances in Environmental Samples. EPA/600/R-93/100 (Washington D.C.).
- Van Metre, P.C., Horowitz, A.J., Mahler, B.J., Foreman, W.T., Fuller, C.C., Burkhardt, M. R., Elrick, K.A., Furlong, E.T., Skrobialowski, S.C., Smith, J.J., Wilson, J.T., 2006. Effects of Hurricanes Katrina and Rita on the chemistry of bottom sediments in Lake Pontchartrain, Louisiana, USA. *Environ. Sci. Technol.* 40 (22), 6894–6902.
- Wang, H., Appan, A., Gulliver, J.S., 2003. Modeling of phosphorus dynamics in aquatic sediments: I—model development. *Water Res.* 37 (16), 3928–3938.
- Wang, H., Jawitz, J.W., White, J.R., Martinez, C.J., Sees, M.D., 2006. Rejuvenating the largest municipal treatment wetland in Florida. *Ecol. Eng.* 26, 132–146.
- Wang, H., Zhang, Z., Liang, D., Pang, Y., Hu, K., Wang, J., 2016. Separation of wind's influence on harmful cyanobacterial blooms. *Water Res.* 98, 280–292.
- Wang, J., Chai, J., Xu, R., Pang, Y., 2024. The effects of wind-wave disturbances on sediment resuspension and phosphate release in Lake Chao. *Sci. Total Environ.* 912, 169254.
- White, J.R., Reddy, K.R., 1999. Influence of nitrate and phosphorus loading on denitrifying enzyme activity in Everglades wetland soils. *Soil Sci. Soc. Am. J.* 63 (6), 1945–1954.
- White, J.R., Fulweiler, R.W., Li, C.Y., Bargu, S., Walker, N.D., Twilley, R.R., Green, S.E., 2009. Mississippi River flood of 2008: observations of a large freshwater diversion on physical, chemical, and biological characteristics of a shallow estuarine lake. *Environ. Sci. Technol.* 43 (15), 5599–5604.
- White, J.R., DeLaune, R.D., Justic, D., Pahl, J., Day, J., Lane, R., Boynton, W., Twilley, R. R., 2019. Consequences of Mississippi River diversions on nutrient dynamics of coastal deltaic wetland soils and estuarine sediments: a review. *Estuar. Coast. Shelf Sci.* 224, 209–216.
- Wright, C.L., Friedrichs, C.T., Massey, G.M., 2022. Controls on sediment bed erodibility in a muddy, partially-mixed tidal estuary. *Front. Earth Sci.* 10, 805130.
- Wu, K., Xu, Y.J., 2007. Long-term freshwater inflow and sediment discharge into Lake Pontchartrain in Louisiana, USA. *Hydrol. Sci. J.* 52 (1), 166–180.
- Wurtsbaugh, W.A., Paerl, H.W., Dodds, W.K., 2019. Nutrients, eutrophication and harmful algal blooms along the freshwater to marine continuum. *Wiley Interdiscip. Rev. Water* 6 (5), e1373.
- Yin, K., Zhang, J., Qian, P.Y., Jian, W., Huang, L., Chen, J., Wu, M.C., 2004. Effect of wind events on phytoplankton blooms in the Pearl River estuary during summer. *Cont. Shelf Res.* 24 (16), 1909–1923.
- You, B.S., Zhong, J.C., Fan, C.X., Wang, T.C., Zhang, L., Ding, S.M., 2007. Effects of hydrodynamics processes on phosphorus fluxes from sediment in large, shallow Taihu Lake. *J. Environ. Sci.* 19 (9), 1055–1060.
- Zhang, W., White, J.R., DeLaune, R.D., 2012. Diverted Mississippi River sediment as a potential phosphorus source affecting coastal Louisiana water quality. *J. Freshw. Ecol.* 47, 575–586.